MAHARSHI OZA

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PROFESSIONAL SUMMARY

RTL Design engineer with 4+ years of experience in digital design and IP verification across ASIC/FPGA and automotive domains. Skilled in synthesizable SystemVerilog/Verilog microarchitecture, FSM and datapath design, pipelining, on-chip memories, and assertion-based design checking. Strong foundation in RISC-V microarchitecture, cache/memory hierarchies, and timing-aware design, with exposure to STA, CDC, and low-power design (IEEE 1801/UPF, clock and power gating).

CORE COMPETENCIES

Languages & Protocols: SystemVerilog, Verilog HDL, C++, Python, Perl, Shell Scripting, AXI4, I2C, RISC-V ISA
Design Verification: UVM, Constrained-Random Verification (CRV), SVA, Functional & Code Coverage, Coverage Closure, Regression Testing, Debug & Root-Cause Analysis, Scoreboards, TLM, SoC/IP Verification
RTL & ASIC: RTL Design, Logic Synthesis, Static Timing Analysis (STA), Clock Domain Crossing (CDC), Low-Power Design (IEEE 1801/UPF, Clock Gating, Power Gating), Design for Testability (DFT)
Tools: VCS, Xcelium, QuestaSim, ModelSim, Design Compiler, Vivado, Git, Linux
Computer Architecture: RISC-V, Pipelining, Superscalar & Out-of-Order Execution, Register Renaming, Branch Prediction, Memory Hierarchy
GPU & AI Hardware: GPU/SIMT Architecture, CUDA, GPU Memory Hierarchy, Warp Execution, Parallel Computing, Matrix Multiplication, Convolution, Parallel Reduction, AI/LLM Accelerators

PROFESSIONAL EXPERIENCE

RTL Design Verification Engineer | Maxgen Technologies Pvt Ltd., India **Aug 2023 – Aug 2024**

- Contributed to the RTL design of a multi-port packet router in Verilog, including FSM-based control, internal FIFOs, synchronizers, and output arbitration; supported synthesis and timing closure in Vivado.
- Implemented an out-of-order scoreboard for an AXI4 memory subsystem in SystemVerilog/UVM, tracking outstanding transactions by ID and checking data and per-ID ordering rules.
- Developed functional coverage achieving over 90% for AXI4 transactions; debugged RTL issues with Questa, Xcelium, VCS, and Vivado, and reviewed RTL changes against specification updates.

Functions Engineer – Vision & Perception | Magna Electronics, India **Dec 2021 – Aug 2023**

- Developed validation code for ADAS 2+ lane-marker detection, checking lane width, inter-lane distance, and related geometry against ground truth, and built KPI evaluation utilities for safety-critical perception features.
- Implemented code, branch, and line coverage for 2 perception modules and the KPI framework to expose validation gaps; root-caused edge-case failures with GDB and applied MISRA C practices in development and review.

PROJECTS

Set-Associative Write-Back Cache Controller | SystemVerilog RTL, SVA

- Designed a parameterized N-way set-associative write-back/write-allocate blocking cache controller with selectable LRU/LFU replacement, per-line dirty tracking, a flush/invalidate engine, and valid/ready CPU and memory interfaces with error propagation.
- Partitioned the design into tag-compare, replacement, dirty-tracking, and single-port SRAM modules under a central FSM; wrote a bound SVA checker for handshake and FSM invariants and verified the design in a UVM environment with 0 data mismatches.

High-Performance Transformer Neural Network with Pipelined Self-Attention | SystemVerilog, Design Compiler

- Designed a pipelined Transformer self-attention accelerator computing Query/Key/Value projections, score and attention matrices from SRAM-stored inputs and weights; passed 217/217 result checks across 4 test cases, closed timing at a 5 ns clock in Nangate 45 nm (~13.7K cells, ~14.7K µm²), achieving an area–performance efficiency of 3.97×10^{-9} $ns^{-1} \cdot \mu m^{-2}$.

FIR Filter Accelerator via High-Level Synthesis | SystemC, Catapult HLS, RISC-V Rocket-Chip

- Implemented a 16-tap FIR filter accelerator in SystemC (FSM-driven, 64-bit packed data channels) and synthesized it with Catapult HLS at a 5 ns clock, working alongside a Rocket-Chip RISC-V system.
- Explored HLS architectures and tracked area, latency and critical path across runs, reducing latency from ~3.5K cycles to 69 cycles at a ~2.8 ns critical path.

Low-Power Design with IEEE 1801 (Cadence course labs) | UPF, Conformal Low Power

- Wrote IEEE 1801 power intent (3 power domains, power switch, isolation cells, 0.9 V → 1.08 V level shifter) and checked it with Conformal Low Power.

Functional Verification of LC-3 Processor | UVM/UVMF

- Architected a reusable UVM/UVMF environment covering LC-3 pipeline stages and instruction/data memory; developed sequences, tests, assertions, regression, and coverage models for control-flow, datapath, and corner cases, achieving 100% functional coverage closure.
- Built independent scoreboard-based checking to validate instruction execution against expected pipeline behavior, enabling systematic bug isolation across fetch, decode, and execute stages.

Stride Value Predictor for RISC-V OOO Pipeline Simulator | C++

- Developed a cycle-accurate RISC-V out-of-order simulator and integrated a Stride Value Predictor with VPQ/checkpoint recovery; added a Global Stride History Table with LRU replacement, improving harmonic-mean IPC from 1.91 to 2.00 across 15 benchmarks within a 27 KB storage budget.

EDUCATION

M.S. Computer Engineering | North Carolina State University, Raleigh, NC | CGPA: 3.7/4 **Aug 2024 – May 2026**
Relevant Coursework: Advanced Verification using UVM, ASIC Verification, ASIC and FPGA Design with Verilog, Advanced Microprocessor Architecture, Hardware–Software Codesign, Physical Design, Microarchitecture Reverse Engineering
Training: Cadence – Fundamentals of IEEE 1801 Low-Power Specification Format (v12.0)

B.Tech. Electronics & Communication Engineering | Dharmsinh Desai University, India **Aug 2018 – May 2022**